

Heaven's Light is Our Guide

Rajshahi University of Engineering & Technology

Department of Computer Science & Engineering



Lab Report

Course Code : CSE 3206
Course Name : Software Engineering Sessional
Date of Submission : 04-08-2026

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Name of the experiment

Software Configuration Management with Git and GitHub.

Objectives

1. To learn the basic use of Git for managing project files.
2. To learn how to create a GitHub repository and connect it with a local project.
3. To understand how to save changes by using Git commit.
4. To learn how to create and use branches for different work.
5. To learn how to push project files to GitHub and pull updated files.
6. To understand how team members can work on the same project by using Git and GitHub.
7. To gain basic practical knowledge about Software Configuration Management using Git and GitHub.

Introduction

Software Configuration Management (SCM) is the process of managing and tracking changes in software during development. It helps developers keep different versions of a project and work together without losing previous work.

Git is a version control system that tracks changes in project files. GitHub is an online platform where Git repositories are stored and shared for collaboration.

Some basic Git commands used in this experiment are:

- **git init** – Creates a new Git repository.
- **git add .** – Adds project files to the staging area.
- **git commit -m "message"** – Saves the changes with a message.
- **git branch** – Creates or displays branches.
- **git checkout branch_name** – Switches to another branch.
- **git merge branch_name** – Merges changes from one branch into another.
- **git push** – Uploads local changes to GitHub.
- **git pull** – Downloads the latest changes from GitHub.

This experiment helped us learn the basic use of Git and GitHub for version control and team collaboration.

Required Equipment / Software

1. A computer with an internet connection.
2. Git installed on your system.
3. A free GitHub account.
4. A text editor or IDE (e.g., VS Code).

Procedure

1. Git was installed from the official Git website, and Git was configured using the user's name and email address. For example:

```
git config --global user.name "Mozahid Islam"  
git config --global user.email "shouravmdmozahidislam@gmail.com"
```

2. The team leader created an empty GitHub repository and invited all team members as collaborators.

3. Each team member accepted the invitation and cloned the shared GitHub repository to their local computer. For example:

```
git clone https://github.com/<team-leader-username>/<repository-name>.git
```

4. The project folder was opened in Git Bash, and the repository status was checked before making any modifications. For example:

```
git status
```

5. Each team member created and switched to a separate feature branch to implement their assigned task. For example:

```
git checkout -b mozahid-feature
```

6. The assigned calculator functionality was implemented by creating or modifying the required project files. For example:

```
calculator_first5.js was updated with the first five calculator functions.
```

7. The modified files were added to the staging area.

For example:

```
git add calculator_first5.js
```

8. The staged changes were committed with an appropriate commit message.

For example:

```
git commit -m "Add first five calculator functions"
```

9. The feature branch was pushed to the team leader's GitHub repository.

For example:

```
git push -u origin mozahid-feature
```

10. After all team members completed their assigned tasks, the team leader reviewed and merged the feature branches into the main branch. A merge conflict was also demonstrated and resolved by manually editing the conflicting file, staging the resolved file, and completing the merge.

11. A public GitHub repository was forked to create a personal copy under the contributor's GitHub account.

12. The forked repository was cloned to the local computer. For example:

```
git clone https://github.com/hana-chan/first-contributions.git
```

13. A new branch was created in the local clone to make the required changes.

For example:

```
git checkout -b my-changes
```

14. A small modification (such as adding a comment or fixing a typo) was made to the project files. The changes were then staged and committed. For example:

```
git add .
```

```
git commit -m "Add comment for Git workflow practice"
```

15. The newly created branch was pushed to the forked repository. For example:

```
git push -u origin my-changes
```

16. Finally, a Pull Request was created from the forked repository to the original repository by providing an appropriate title and description, demonstrating the standard GitHub contribution workflow.

Note: In this experiment, each team member worked on a separate feature branch of the team leader's GitHub repository to implement their assigned tasks. After completing the work, each member pushed their respective branch and created a Pull Request. The team leader then reviewed and merged the completed feature branches into the main branch. Additionally, the forking and pull request workflow was demonstrated by forking a public repository, making changes in a separate branch, and submitting a Pull Request to the original repository.

Commands

Repository Initialization and GitHub Setup

```
G:\3-2\CSE 3205\LAB\Calculator-Testing>git push -u origin mozahid-feature
To https://github.com/jabin-chan/Calculator-Testing.git
```

```
Git CMD
G:\>cd "G:\3-2\CSE 3205\LAB"

G:\3-2\CSE 3205\LAB>git clone https://github.com/jabin-chan/Calculator-Testing.git
fatal: destination path 'Calculator-Testing' already exists and is not an empty directory.

G:\3-2\CSE 3205\LAB>git clone https://github.com/jabin-chan/Calculator-Testing.git
Cloning into 'Calculator-Testing'...
remote: Enumerating objects: 43, done.
remote: Counting objects: 100% (43/43), done.
remote: Compressing objects: 100% (35/35), done.
remote: Total 43 (delta 4), reused 41 (delta 2), pack-reused 0 (from 0)
Receiving objects: 100% (43/43), 84.48 KiB | 318.00 KiB/s, done.
Resolving deltas: 100% (4/4), done.

G:\3-2\CSE 3205\LAB>cd Calculator-Testing

G:\3-2\CSE 3205\LAB\Calculator-Testing>git checkout -b mozahid-feature
Switched to a new branch 'mozahid-feature'

G:\3-2\CSE 3205\LAB\Calculator-Testing>copy "G:\3-2\CSE 3205\LAB\calculator_first5.js" "G:\3-2\CSE 3205\Calculator-Testing\"
Overwrite G:\3-2\CSE 3205\Calculator-Testing\calculator_first5.js? (Yes/No/All): Yes
1 file(s) copied.

G:\3-2\CSE 3205\LAB\Calculator-Testing>git status
On branch mozahid-feature
nothing to commit, working tree clean

G:\3-2\CSE 3205\LAB\Calculator-Testing>git add calculator_first5.js
fatal: pathspec 'calculator_first5.js' did not match any files

G:\3-2\CSE 3205\LAB\Calculator-Testing>git commit -m "Add first 5 calculator functions"
On branch mozahid-feature
nothing to commit, working tree clean
```

Setup from another member's local device

```
● (base) hanalala@Jabins-Laptop functionlast5 % git init
Reinitialized existing Git repository in /Users/hanalala/Desktop/swetest/functionlast5/.git/
● (base) hanalala@Jabins-Laptop functionlast5 % git add .
● (base) hanalala@Jabins-Laptop functionlast5 % git commit -m "fixing last 5"
[last-5-functions 14e50d5] fixing last 5
1 file changed, 1 insertion(+), 1 deletion(-)
```

Branching and Merging

```
● (base) hanalala@Jabins-Laptop functionlast5 % git branch
* last-5-functions
  main
● (base) hanalala@Jabins-Laptop functionlast5 % git fetch -a
remote: Enumerating objects: 38, done.
remote: Counting objects: 100% (38/38), done.
remote: Compressing objects: 100% (32/32), done.
remote: Total 37 (delta 2), reused 37 (delta 2), pack-reused 0 (from 0)
Unpacking objects: 100% (37/37), 83.88 KiB | 584.00 KiB/s, done.
From https://github.com/jabin-chan/Calculator-Testing
 * [new branch]      mozahid-feature -> origin/mozahid-feature
 * [new branch]      mursalin-frontend -> origin/mursalin-frontend
● (base) hanalala@Jabins-Laptop functionlast5 % git branch -a
* last-5-functions
  main
remotes/origin/HEAD -> origin/last-5-functions
remotes/origin/last-5-functions
remotes/origin/mozahid-feature
remotes/origin/mursalin-frontend
● (base) hanalala@Jabins-Laptop functionlast5 % git checkout -b mozahid-feature origin/mozahid-feature
git checkout -b mursalin-frontend origin/mursalin-frontend
branch 'mozahid-feature' set up to track 'origin/mozahid-feature'.
Switched to a new branch 'mozahid-feature'
branch 'mursalin-frontend' set up to track 'origin/mursalin-frontend'.
Switched to a new branch 'mursalin-frontend'
```

```

7267488 (main) trying
(base) hanalala@Jabins-Laptop functionlast5 % git checkout main
Switched to branch 'main'
(base) hanalala@Jabins-Laptop functionlast5 % git branch
  last-5-functions
* main
  mozahid-feature
  mursalin-frontend
(base) hanalala@Jabins-Laptop functionlast5 % git merge last-5-functions
Updating 7267488..14e50d5
Fast-forward
 aa.js      | 0
 last5.js   | 53 ++++++++++++++++++++++++++++++++++++++++++++++++++++++
 2 files changed, 53 insertions(+)
 delete mode 100644 aa.js
 create mode 100644 last5.js
⊗ (base) hanalala@Jabins-Laptop functionlast5 % git merge mozahid-feature
error: there was a problem with the editor 'vi'
Not committing merge; use 'git commit' to complete the merge.
● (base) hanalala@Jabins-Laptop functionlast5 % git commit -m "Merge branch 'mozahid-feature'"
[main c053309] Merge branch 'mozahid-feature'

```

Merge Conflict

```

⊗ (base) hanalala@Jabins-Laptop functionlast5 % git merge mursalin-frontend
Auto-merging Calculator_frontend/src/last5.js
CONFLICT (content): Merge conflict in Calculator_frontend/src/last5.js
Automatic merge failed; fix conflicts and then commit the result.
● (base) hanalala@Jabins-Laptop functionlast5 % git status
On branch main
You have unmerged paths.
  (fix conflicts and run "git commit")
  (use "git merge --abort" to abort the merge)

Changes to be committed:
  new file:   Calculator_frontend/.gitignore
  new file:   Calculator_frontend/README.md
  new file:   Calculator_frontend/components.json
  new file:   Calculator_frontend/eslint.config.js
  new file:   Calculator_frontend/index.html
  new file:   Calculator_frontend/jsconfig.json
  new file:   Calculator_frontend/package-lock.json
  new file:   Calculator_frontend/package.json
  new file:   Calculator_frontend/public/favicon.svg
  new file:   Calculator_frontend/public/icons.svg
  new file:   Calculator_frontend/src/App.css
  new file:   Calculator_frontend/src/App.jsx
  new file:   Calculator_frontend/src/assets/hero.png
  new file:   Calculator_frontend/src/assets/react.svg
  new file:   Calculator_frontend/src/assets/vite.svg
  new file:   Calculator_frontend/src/calculator_first5.js
  renamed:   calculator_logic.js -> Calculator_frontend/src/calculator_logic.js
  new file:   Calculator_frontend/src/components/ui/button.jsx
  new file:   Calculator_frontend/src/components/ui/hacker-background.jsx
  new file:   Calculator_frontend/src/index.css
  new file:   Calculator_frontend/src/lib/Utils.js
  new file:   Calculator_frontend/src/main.jsx
  new file:   Calculator_frontend/vite.config.js
  deleted:   calculator_first5.js
  deleted:   last5.js

Unmerged paths:
  (use "git add <file>..." to mark resolution)
  both modified: Calculator_frontend/src/last5.js

```


Resolving Merge Conflicts

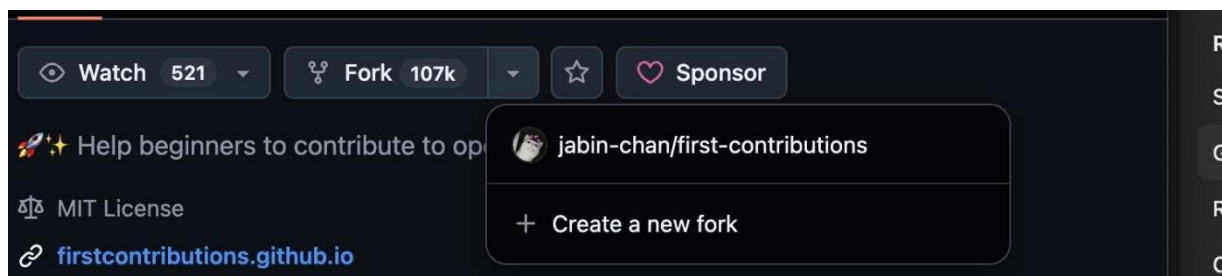
```
Calculator_frontend > src > JS last5.js > ...
You, 9 minutes ago | 2 authors (You and one other) | Accept Current Change | Accept Incoming Change | Accept Both Changes | Compare Changes
<<<<<< HEAD:last5.js (Current Change)      You, 9 minutes ago * Uncommitted changes
function squareRoot(num) {
  // Square Root
  export function squareRoot(num) {
  >>>>>> mursalin-frontend:Calculator_frontend/src/last5.js (Incoming Change)

(base) hanalala@Jabins-Laptop Calculator_frontend % cd ..
(base) hanalala@Jabins-Laptop functionlast5 % git add Calculator_frontend/src/last5.js
(base) hanalala@Jabins-Laptop functionlast5 % git status
On branch main
All conflicts fixed but you are still merging.
(use "git commit" to conclude merge)

Changes to be committed:
  new file:   Calculator_frontend/.gitignore
  new file:   Calculator_frontend/README.md
  new file:   Calculator_frontend/components.json
  new file:   Calculator_frontend/eslint.config.js
  new file:   Calculator_frontend/index.html
  new file:   Calculator_frontend/jsconfig.json
  new file:   Calculator_frontend/package-lock.json
  new file:   Calculator_frontend/package.json
  new file:   Calculator_frontend/public/favicon.svg
  new file:   Calculator_frontend/public/icons.svg
  new file:   Calculator_frontend/src/App.css
  new file:   Calculator_frontend/src/App.jsx
  new file:   Calculator_frontend/src/assets/hero.png
  new file:   Calculator_frontend/src/assets/react.svg
  new file:   Calculator_frontend/src/assets/vite.svg
  new file:   Calculator_frontend/src/calculator_first5.js
  renamed:   calculator_logic.js -> Calculator_frontend/src/calculator_logic.js
  new file:   Calculator_frontend/src/components/ui/button.jsx
  new file:   Calculator_frontend/src/components/ui/hacker-background.jsx
  new file:   Calculator_frontend/src/index.css
  renamed:   last5.js -> Calculator_frontend/src/last5.js
  new file:   Calculator_frontend/src/lib/Utils.js
  new file:   Calculator_frontend/src/main.jsx
  new file:   Calculator_frontend/vite.config.js
  deleted:   calculator_first5.js

(base) hanalala@Jabins-Laptop functionlast5 % git commit -m "Fix: Resolve merge conflict"
[main 9f7f4fa] Fix: Resolve merge conflict
```

Forking and Pull Request



```
fatal: repository 'https://github.com/hanalala/first-contributions.git' not found
(base) hanalala@Jabins-Laptop swetest % git clone https://github.com/jabin-chan/first-contributions.git
Cloning into 'first-contributions'...
remote: Enumerating objects: 90191, done.
remote: Counting objects: 100% (132/132), done.
remote: Compressing objects: 100% (65/65), done.
remote: Total 90191 (delta 106), reused 70 (delta 67), pack-reused 90059 (from 3)
Receiving objects: 100% (90191/90191), 38.13 MiB | 5.21 MiB/s, done.
Resolving deltas: 100% (58900/58900), done.
```

```
(base) hanalala@Jabins-Laptop swetest % cd first-contributions
(base) hanalala@Jabins-Laptop first-contributions % git checkout -b my-changes
Switched to a new branch 'my-changes'
(base) hanalala@Jabins-Laptop first-contributions % git add .
(base) hanalala@Jabins-Laptop first-contributions % git commit -m "done"
[my-changes 98be9a2c3] done
1 file changed, 1 insertion(+)
(base) hanalala@Jabins-Laptop first-contributions %
```

 Public repository · Forked from [firstcontributions/first-contributions](#)

 my-changes had recent pushes 1 minute ago

Compare & pull request

 main ▾

Go to file

+

 Code ▾

Output

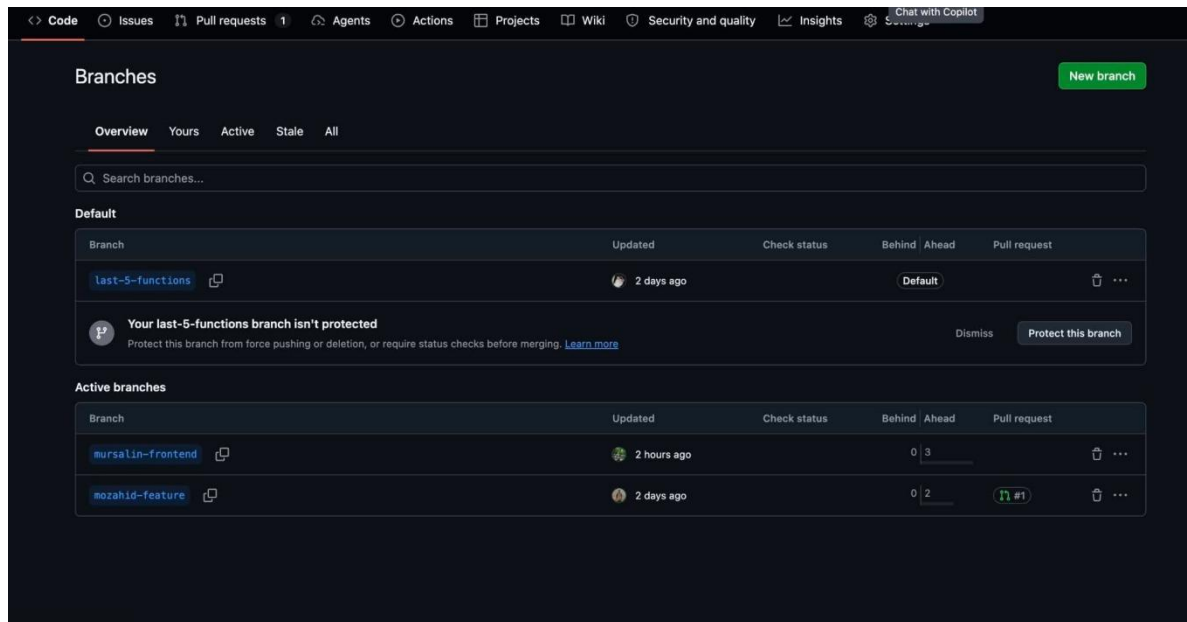


Fig.03 GitHub repository

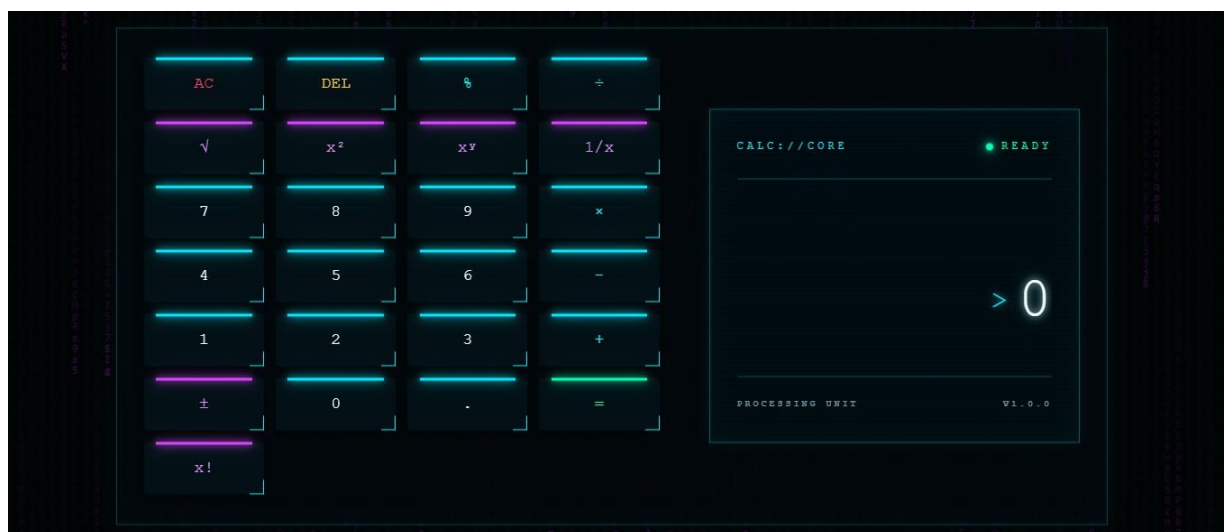


Fig.04 Project Output

Precautions

1. Configure the Git user name and email before making the first commit.
2. Verify that the correct branch is selected before making changes or pushing code.
3. Use clear and meaningful commit messages.
4. Review staged files before committing.
5. Never commit sensitive information such as passwords or API keys.
6. Resolve merge conflicts carefully before completing a merge.

Conclusion

This experiment provided practical experience with Git and GitHub for version control and collaborative software development. Through repository management, branching, merging, conflict resolution, and the fork-and-pull-request workflow, the team gained a better understanding of how developers collaborate efficiently while maintaining a clean and organized project history.